

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
        perror("Execution Failed");
    }
    else if(pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }
    return 0;
}

```

Try uname, lscpu, lsblk, ps, top commands after this program

OUTPUT:

```

Last login: Tue Aug 11 09:32:20 on ttys000
eeshayuktha@Eeshas-MacBook-Air ~ % cd ~/Documents/OSLab
eeshayuktha@Eeshas-MacBook-Air OSLab % ./hi
Enter Linux command: uname
Parent PID : 20240
Child PID : 20241
Darwin
Child process completed.
eeshayuktha@Eeshas-MacBook-Air OSLab % ./hi
Enter Linux command: lscpu
Parent PID : 20242
Child PID : 20245
Execution Failed: No such file or directory
Child process completed.
eeshayuktha@Eeshas-MacBook-Air OSLab % ./hi
Enter Linux command: lsblk
Parent PID : 20249
Child PID : 20252
Execution Failed: No such file or directory
Child process completed.
eeshayuktha@Eeshas-MacBook-Air OSLab % ./hi
Enter Linux command: ps
Parent PID : 20256
Child PID : 20258
  PID TTY          TIME CMD
 20232 ttys000    0:00.03 -zsh
 20256 ttys000    0:00.01 ./hi
Child process completed.
eeshayuktha@Eeshas-MacBook-Air OSLab %

```

2. copying the contents from one file to another file

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int src, dest;
    char buffer[1024];
    int bytes;

    src = open("input.txt", O_RDONLY);

    if(src < 0)
    {
        printf("Cannot open source file\n");
        return 1;
    }

    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    while((bytes = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, bytes);
    }

    close(src);
    close(dest);

    printf("File copied successfully.\n");

    return 0;
}
```

OUTPUT:

```
eeshayuktha@Eeshas-MacBook-Air ~ % cd ~/Documents/OSLab
eeshayuktha@Eeshas-MacBook-Air OSLab % nano filecopy.c
eeshayuktha@Eeshas-MacBook-Air OSLab % gcc filecopy.c -o file
eeshayuktha@Eeshas-MacBook-Air OSLab % ./file
Cannot open source file
eeshayuktha@Eeshas-MacBook-Air OSLab %
```